DATA SCIENCE XGBoost Model

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XGBoost Model

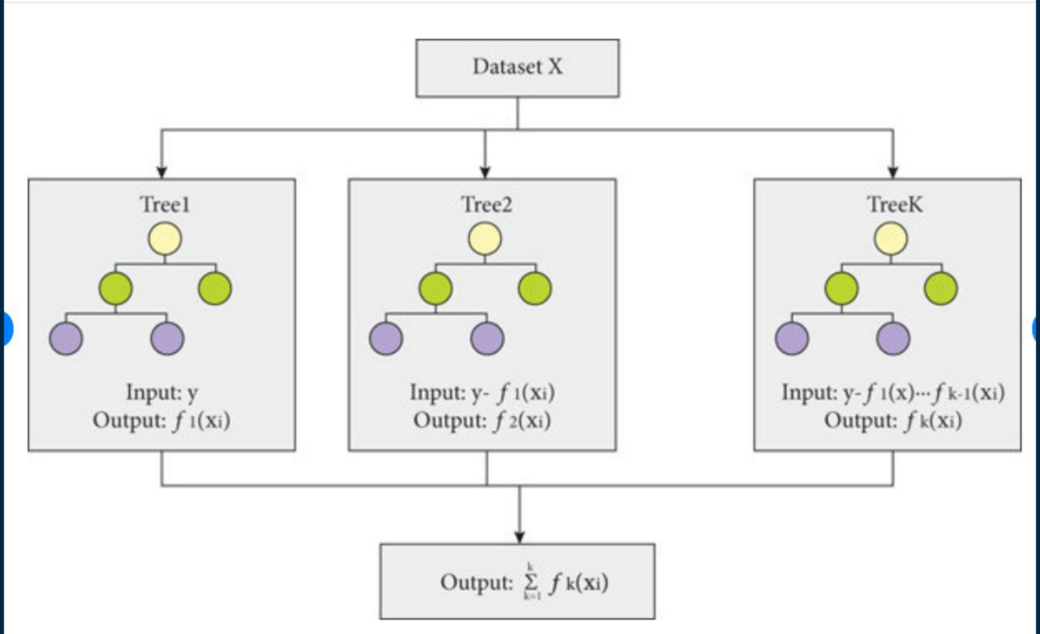
XGBoost (eXtreme Gradient Boosting) is a machine learning library that implements an optimized and distributed Gradient Boosting machine learning algorithm.

The XGBoost Model is can be used to replace logistic regression. It is more robust than logistic regression.

XGBoost is one of the most commonly used models when predicting data in the industry.

XGBoost Framework

It is an **ensemble** of weak prediction models, i.e., models that make very few assumptions about the data, which are typically **simple decision trees**.



It makes decisions based on many decision trees.

XGBoost vs Logistic Regression

	XGBoost	Logistic Regression
Interpretability	Good	Fair
Performance in a dataset with a linear relationship	Fair	Good
Performance in a large number of features	Good	Poor
Performance in a small training dataset	Good	Poor
Performance in a dataset with lot of outlier	Good	Poor (or remove the outlier from the dataset)
Performance in a skewed dataset	Good	Poor (or rebalance weight to the minor)
Performance in a continuous dataset	Bad	Good
Missing values handling in dataset	Good	Poor (or remove/patch the missing values from the dataset)
Ease of Decision Making	Automatically handles	Threshold can be set
Commonality	Rapidly trending up	Widely adopted

XGBoost Parameters

- eta (learning rate): The learning rate controls the step size at which the optimizer makes updates to the weights.
- Max_depth: The max_depth parameter controls the maximum depth of the trees in the model. A larger max_depth value results in more complex models, which can lead to overfitting.
- Subsample: The subsample parameter controls the fraction of observations used for each tree. A smaller subsample value results in smaller and less complex models, which can help prevent overfitting.

XGBoost Parameters – cont.

- Colsample_bytree: The colsample_bytree parameter controls the fraction of features used for each tree. A smaller colsample_bytree value results in smaller and less complex models, which can help prevent overfitting.
- Lambda: The lambda parameter is the L2 regularization term on weights. Larger values means more conservative model, it helps to reduce overfitting by adding a penalty term to the loss function.
- Alpha: The alpha parameter is the L1 regularization term on weights. Larger values means more conservative model, it helps to reduce overfitting by adding a penalty term to the loss function.
- N_estimators: The n_estimators parameter controls the number of trees in the model. Increasing this value generally improves model performance, but can also lead to overfitting. A common value for this parameter is between 100 and 1000.

Case Study - Diabete Prediction

- See handouts
- Notebook:

  https://colab.research.google.com/drive/1XsRIBE7a-IaWUsRI9DM9mZokK1hC3P3B?authuser=1#scrollTo=wMGoo_jJ8neZ

- Dataset: https://drive.google.com/file/d/17ATVGXyP0o1Hj-IRjUBSm33YCYadzD0r/view?usp=drive_link